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Toxicant default guideline values for aquatic ecosystem protection

Nickel in marine water

Technical brief

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Cataloguing data

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Summary

Nickel is a commonly occurring natural element that is essential to some organisms. Nickel is mined and processed globally and used for many purposes, including the production of alloys, food preparation equipment, mobile telephones, batteries, medical equipment, transport, buildings and power generation. Anthropogenic sources of nickel include motor vehicle emissions, landfills, sewage, stormwater runoff and industries such as mining.

The ANZECC/ARMCANZ (2000a) nickel default guideline value (DGV) for 95% species protection in marine water was a 'high reliability' value of 70 µg/L, based on chronic toxicity data for 15 species from five taxonomic groups. However, the 99% species protection value of 7 µg/L was recommended for application to slightly-to-moderately disturbed ecosystems because the 95% species protection DGV was potentially not protective of chronic toxicity to key species. Since 2000, more toxicity data have become available, including data for tropical and temperate organisms and for Australian and/or New Zealand species, from which updated DGVs have been derived. The DGVs reported in this technical brief are based on the guideline values derived by Gissi et al. (2020).

DGVs for nickel in marine water were derived using chronic toxicity data for 17 temperate species and 14 tropical species (combined dataset of 31 species representing 15 taxonomic groups). Using the (shiny)ssdtools software, the fit of the species sensitivity distribution (SSD) to the toxicity data was good, resulting in very high reliability DGVs. The DGVs for 99%, 95%, 90% and 80% species protection are 1.4 µg/L, 5.9 µg/L, 12 µg/L and 27 µg/L, respectively. The DGVs relate to dissolved nickel, operationally defined as the < 0.45-µm filtered measurement. The 95% species protection DGV is recommended for application to slightly-to-moderately disturbed ecosystems.

The DGVs can be used for both temperate and tropical ecosystems, and supersede the ANZECC/ARMCANZ (2000) DGVs for nickel in marine water.

1 Introduction

Nickel is the fifth most common element on earth and occurs expansively in the earth's crust (Nickel Institute 2015). It primarily occurs as oxides, sulfides and silicates (Pyle and Couture 2012). Nickel ores are mined in over 23 countries and are smelted or refined in 25 countries, including Australia. Approximately 2.5 million tonnes of nickel are produced annually, and the world demand for nickel was growing at an average rate of 5% per annum over 2010–2020 (INSG 2024).

More than two-thirds of nickel produced is used in the production of alloys (e.g. stainless steel) with other metals such as iron, copper and chromium (INSG 2024). Nickel-containing materials include food preparation equipment, mobile telephones, medical equipment, automotive and engine components, buildings, batteries and power generation (Nickel Institute 2015). Anthropogenic sources of nickel include motor vehicle emissions, landfills, sewage, stormwater runoff and industries such as mining. Magmatic sulfide and laterite ores are naturally enriched in nickel. Nickel laterites have a fine dispersive nature and are formed by the extensive chemical and physical weathering of ultramafic rocks under tropical, humid conditions (Mudd 2010). Recent estimates show that 60% of the world's nickel reserves are contained in laterite deposits (USGS 2019). In 2018, 48% of the world's nickel production came from the tropical Asia–Pacific region, including Indonesia, New Caledonia and the Philippines (USGS 2019).

Nickel predominantly occurs in the +2 oxidation state (i.e. Ni^{2+}) and forms stable complexes with inorganic and organic ligands (Eisler 1998; Pyle and Couture 2012). In seawater, Ni^{2+} is the main form of nickel (~36%), followed by chloride (27%) and carbonate (19%) species of nickel (Kumar 1986). Once nickel has entered an aquatic system, it can be accumulated by biota (e.g. phytoplankton, aquatic plants) or it can be deposited in the sediment by precipitation, complexation and adsorption on clay particles, with subsequent uptake in benthic biota (Cempel and Nikel 2006).

Concentrations of nickel in unimpacted marine coastal water using ultratrace sampling and analysis are typically <0.2 µg/L (Apte et al. 2018). Older data on background concentrations of nickel in seawater sampled in the North Pacific ranged from 0.15 µg/L to 0.66 µg/L (Bruland 1980). In Europe, Heijerick and Van Sprang (2008) reported 3.3 µg/L and 0.3 µg/L as the highest nickel concentrations for estuarine/coastal water and open ocean water, respectively. Van Geen and Luoma (1993) demonstrated that dissolved nickel concentrations increase closer to shore, with concentrations offshore of San Francisco Bay ranging from 0.26 µg/L to 0.32 µg/L and concentrations nearer to shore of ≤0.94 µg/L.

In some regions, such as New Caledonia, nickel concentrations in soils and aquatic systems are naturally enriched, but mining lateritic nickel ores can increase the input of metals into the coastal system. Dissolved nickel concentrations in New Caledonian seawater have been reported to range from <0.1 µg/L to 11 µg/L (Hedouin et al. 2009).

Nickel is an essential nutrient for micro-organisms and terrestrial plants, and at least eight nickel-containing enzymes have been identified (Moreton et al. 2009). In aquatic plants and cyanobacteria, the necessity of nickel has been documented in urease and hydrogenase metabolism; however, its necessity in aquatic animals has not been confirmed (Muyssen et al. 2004).

The previous default guideline values (DGVs) for nickel in marine water were derived from chronic toxicity data for 15 species from five taxonomic groups (ANZECC/ARMCANZ 2000a). The 99% and 95% species protection DGVs were 7 µg/L and 70 µg/L, respectively. The 99% species protection DGV was recommended for application to slightly-to-moderately disturbed marine systems because the 95% species protection value did not sufficiently protect the following organisms: a group of species with acute toxicity values close to the 95% DGV; a juvenile mysid (152 µg/L (Gentile et al. 1982)); and unconfirmed data for a mollusc (61 µg/L), a diatom (50–100 µg/L) and two dinoflagellates (100 µg/L).

Since the derivation of the ANZECC/ARMCANZ (2000a) DGVs for nickel in marine water, DeForest and Schlekat (2013) derived a 5% hazardous concentration (HC5; analogous to a 95% species protection value) for nickel in marine water of 3.9 µg/L, largely influenced by data for the highly sensitive tropical sea urchin *Diadema antillarum* (EC10 of 2.9 µg/L), which is endemic to the Caribbean Sea (Bielmyer et al. 2005), although *Diadema* species have recently been found in the Indo-Pacific. If this species was excluded, the HC5 was 21 µg/L. DeForest and Schlekat (2013) recommended that *D. antillarum* be excluded from the temperate dataset for European waters because it was not endemic and because local urchin species were included. Notably, under the ANZG (2018) Guidelines DGV derivation method (Warne et al. 2018), there was no basis upon which to exclude this species from the dataset used to derive the current DGVs.

Since 2000, more toxicity data for nickel in marine water have become available, including data for tropical, temperate and local species, from which updated DGVs have been derived. The DGVs reported here are based on the nickel marine guideline values derived by Gissi et al. (2020). They supersede the ANZECC/ARMCANZ (2000a) DGVs for nickel in marine water.

2 Aquatic toxicology

2.1 Mechanism of toxicity

Brix et al. (2017) provided a comprehensive review of the current understanding of the mechanisms of nickel toxicity to aquatic biota, although the focus was on freshwater organisms and some of the mechanisms may not be relevant to marine organisms. In marine and estuarine water, factors such as water chemistry and the physiology of estuarine and marine biota are expected to alter the mechanisms of toxicity and toxicological impact (Blewett and Leonard 2017). Mechanisms reported by Brix et al. (2017) include disruption of calcium, magnesium and iron homeostasis, induced oxidative damage via reactive oxygen species, and an allergic response of respiratory epithelia. The reduced calcium availability is known to affect exoskeleton, shell and bone growth in invertebrates (Brix et al. 2017). For aquatic plants, in addition to oxidative damage, high concentrations of nickel may displace magnesium from the chlorophyll molecule and inhibit photosynthesis (Brix et al. 2017).

Blewett and Leonard (2017) also reported ionoregulatory impairment, inhibition of respiration, and promotion of oxidative stress as the three main mechanisms of toxicity in marine invertebrates and fish. They concluded that, despite changes in the speciation of nickel in marine water, organism physiology appeared to be the key driver of toxic impact (Blewett and Leonard 2017).

Evidence of these effects on aquatic biota in chronic exposures at nickel concentrations found in the environment is limited. There has also been a lack of studies of diverse taxonomic groups and tropical species (Blewett and Leonard 2017). As such, the mechanisms of nickel toxicity in marine water are not well understood.

2.2 Toxicity

Over the past two decades, the body of literature on the toxicity of nickel to marine species has increased greatly, with the dataset spanning over 40 species from over 15 taxonomic groups and representing temperate and tropical species. As reported in Section 4.1 and by Gissi et al. (2020), there appears to be no significant difference in the toxicity of nickel between temperate and tropical species.

Echinoderm (sea urchin) early life stages were the most sensitive species to nickel. *Evichinus chloroticus* had a 96-h EC50 (larval abnormality) of 14 µg/L (Blewett et al. 2016), while *Diadema antillarum* had a 40-h EC10 (larval abnormality) of 2.9 µg/L (Deforest and Schlekat 2013). However, two species of echinoderm, *Dendraster excentricus* and *Strongylocentrotus purpuratus*, showed markedly lower sensitivity than other echinoderms, with 48-h EC10s (larval abnormality) of 191 µg/L and 335 µg/L, respectively. For sea urchin data, in two cases where both chronic EC50 and NOEC/EC10 values were available, factors of 5.2 and 5.0 were found for the EC50:NOEC/EC10 ratio, which supports the default conversion factor of 5 applied to chronic EC50 data to estimate negligible effect values as part of the Warne et al. (2018) derivation method.

Crustaceans and gastropods also showed high sensitivity to nickel. The tropical copepod *Acartia sinjiensis* was very sensitive, with an 80-h EC10 (development) of 5.5 µg/L (Gissi et al. 2018), while a 28-d NOEC (mortality) of 10 µg/L was reported for the mysid shrimp *Mysidopsis intii* (Hunt et al. 2002). Chronic EC10s or NOECs (for various endpoints and durations) of approximately 22–94 µg/L have been reported for the gastropods *Haliotis rufescens*, *Nassarius dorsatus* and *Monodonta labio* (Hunt et al. 2002; Gissi et al. 2018; Wang et al. 2019).

Bivalves and macroalgae appear to show intermediate sensitivity to nickel. Reported 48–96-h EC10/EC20s for bivalves range from approximately 90 µg/L to 430 µg/L, while 10-d EC10 values for macroalgae ranged from approximately 100 µg/L to 150 µg/L. Algal species (including green algae, diatoms, cyanobacteria) are generally insensitive to nickel, with EC10/NOEC values ranging from 90 µg/L to 17 900 µg/L, while corals and fish are also amongst the least sensitive taxa, with EC10/NOEC values ranging from approximately 900 µg/L to 2 000 µg/L and from approximately 3 000 µg/L to 20 000 µg/L, respectively.

3 Factors affecting toxicity

The dissolved form of nickel, particularly the free cation Ni^{2+}, is the most toxic form of nickel. Increased salinity lowers the concentration of Ni^{2+} due to complexation with chloride (Byrne 2002). These changes in nickel speciation with differing pH and salinity affect its toxicity to aquatic organisms. Typically, bioavailability and toxicity decrease as salinity increases (Hall and Anderson 1995, Blewett et al. 2015).

The water chemistry component most relevant to nickel bioavailability in marine water is dissolved organic carbon (DOC), as pH and cation content (salinity) do not vary substantially among marine sites[1]. The effects of DOC on nickel toxicity are less clear for marine water than for freshwater. No clear influence of DOC was seen for the mussel *Mytilus galloprovincialis* for DOC in the range 1.2–2.7 mg/L or for the diatom *Selenastrum costatum* in the range 0.2–2.7 mg/L (Deforest and Schlekat 2013). Blewett et al. (2016, 2018) showed that nickel toxicity to the urchin *E. chloroticus* and the mussel *Mytilus edulis* was influenced by both DOC quantity and quality. However, nickel toxicity varied by less than a factor of 2 among different natural water sources. Absorbance at 340 nm, which is indicative of the humic/fulvic content of the DOC, showed the strongest relationship with amelioration of nickel toxicity by DOC (Blewett et al. 2016, 2018). Due to the minimal effect DOC has been observed to have on nickel toxicity, no bioavailability normalisation approach has been recommended for the nickel DGVs in marine water.

4 Default guideline value derivation

The DGVs were derived in accordance with the method described in Warne et al. (2018) except that the shinyssdtools software (V 0.4.0) (Dalgarno 2018) was used instead of the Burrlioz 2.0 software. This software has now superseded the use of Burrlioz 2.0 software as used by Gissi et al. (2020).

4.1 Toxicity data used in derivation

The derivation of DGVs for nickel in marine water relied heavily on the nickel in marine water guideline values derived by Gissi et al. (2020). However, the process also involved searching the international literature for published studies that assessed the toxicity of nickel in marine water. Data from these studies were then collated, screened and analysed according to the methods described by Warne et al. (2018).

Due to the large size of the nickel toxicity dataset, it was initially divided into data for temperate species and data for tropical species. Temperate biota were defined as species isolated from temperate regions and/or having a natural geographical distribution outside of the Tropics of Cancer and Capricorn, and toxicity tests were conducted at temperatures <25°C. Tropical biota were defined as species that have a natural geographical distribution between the Tropic of Cancer and the Tropic of Capricorn, and toxicity tests were conducted at ≥25°C. Tests had measured salinity of ≥25 ppt, although one species, *Artemia salina*, was tested in seawater but salinity was not measured.

A comprehensive comparison of the temperate and tropical datasets (Gissi et al. 2020) concluded the DGVs should be derived from the combined dataset of both temperate and tropical species. Briefly, the sensitivities of temperate and tropical marine species to nickel were similar, and species sensitivity distributions (SSDs) based on the temperate and tropical datasets resulted in protective concentrations (for 80%, 90%, 95% and 99% species protection) for temperate and tropical species that were not significantly different from each other (see Gissi et al. 2020). Therefore, there is no need for separate temperate and tropical DGVs for nickel in marine water. Moreover, combining the

[1] A more detailed description of factors affecting nickel toxicity is provided in the nickel in freshwater DGVs technical brief (ANZG 2025).

temperate and tropical datasets results in a larger dataset that includes more taxonomic groups (Gissi et al. 2020), which ultimately improves the confidence in the DGVs.

Measurements of DOC were not always reported in the original toxicity studies, but where reported were typically lower than 2.7 mg/L, where negligible amelioration of nickel toxicity would be expected.

In cases where both NOEC and EC10 (or similar) values were available for the same species, professional judgement was used to select the most appropriate value. In making the decisions, the concentration–response relationships were closely examined to determine the toxicity value that best represented a negligible effect concentration. For the crustacean *Mysidopsis intii*, a NOEC of 10 µg/L for survival was chosen over an EC10 of 45.2 µg/L for growth (Hunt et al. 2002), and for the gastropod *Haliotis rufescens*, a NOEC of 21.5 µg/L for shell growth was chosen over an EC10 of 36.4 µg/L for metamorphosis (Hunt et al. 2002). For the brown-golden alga *Tisochrysis lutea*, a NOEC of 250 µg/L for 72-h growth rate was chosen over an EC10 of 330 µg/L (Gissi 2018). In the case of the coral *Acropora digitifera,* an EC5 of 1680 µg/L for 5-h fertilisation success was chosen over a NOEC of 940 µg/L (Gissi et al. 2017) as the concentration–response relationship suggested that the latter value was highly conservative.

The acceptable quality chronic toxicity dataset totalled 42 species from 15 taxonomic groups, and included chronic EC5, EC10, EC20, NOEC, LOEC, MATC and EC50 values. The dataset included nine values that needed to be converted to negligible effect equivalents if they were going to be used for the derivation (i.e. one MATC, one LOEC and seven EC50s). Given the high number of negligible effect values that were in the dataset (i.e. 33), a decision was made to not include the converted MATC, LOEC and EC50 values, resulting in a dataset of 33 species for 14 taxonomic groups. The effect of this on the derived protective concentration (PC) values is discussed in Appendix B.

Two of the NOEC values were reported as “less than” values (i.e. <280 µg/L for the coral, *Acropora aspera* (Gissi et al. 2017) and <10 µg/L for the echinoderm *Hemicentrotus pulcherrimus* (Hwang et al., 2012). Although Warne et al. (2018) allows such data to be used under specific circumstances, inclusion of the values had only a very minor effect on the DGVs (see Appendix B) and, given the uncertainty of the values, they were excluded from the final dataset, resulting in a dataset of 31 species for 14 taxonomic groups.

The 31 remaining acceptable quality chronic toxicity values included eight NOECs and an EC20 in addition to 20 EC10s, one EC5 and one LC10. Warne et al. (2018) state that NOECs are not preferred estimates of toxicity and their inclusion in datasets where n > 8 needs to be justified. The same requirement would logically extend to EC20s. Therefore, the effect of excluding the NOEC and EC20 values on the DGVs was assessed (see Appendix B). The conclusion was that, although there were some differences in the DGVs between the datasets, there was an overall net benefit of including the NOEC and EC20 values. Consequently, the final dataset that was used to derive the DGVs comprised toxicity data for 31 species from 14 taxonomic groups, and included 17 temperate species from 10 taxonomic groups and 14 tropical species from 10 taxonomic groups. A summary of the toxicity data used to derive the DGVs is in Table 1. Further details about the data and test conditions are in Appendix A. Details of the data quality assessment and the data that passed the quality assessment are provided as supporting information.

An assessment of the modality of the final dataset confirmed that the dataset was not bimodal or multimodal as shown in Appendix C.

Table 1 Summary of single chronic toxicity values, all species used to derive the default guideline values for nickel in marine water

Taxonomic group	Species	Life stage	Duration	Toxicity measure [a] (endpoint)	Final toxicity value (µg/L)
Cyanobacterium	*Cyanobium* sp.	6 x 10^3 cells/mL	72 h	EC10 (growth rate)	3 700
Diatom	*Ceratoneis closterium*	5–6 d old, 1–3 x 10^3 cells/mL	72 h	EC10 (growth rate)	2 870 [a]
	Skeletonema costatum	–	96 h	EC10 (growth)	132 [a]
Green alga	*Dunaliella tertiolecta*	–	96 h	EC10 (growth)	17 900
Brown-golden alga	*Tisochrysis lutea*	5–6 d old, 1–3 x 10^3 cells/mL	72 h	NOEC (growth rate)	250
Dinoflagellate	*Symbiodinium* sp. Freud. Clade C.	6–7 d old, 1–3 x 10^3 cells/mL	72 h	NOEC (growth rate)	310
Red macroalga	*Champia parvula*	Adult	10 d	EC10 (reproduction)	144
Brown macroalga	*Macrocystis pyrifera*	Zoospore	10 d	EC10 (reproduction)	96.7
Crustacean	*Acartia sinjiensis*	Egg	80 h	EC10 (development)	5.5
	Amphibalanus amphitrite	Nauplius	96 h	EC10 (metamorphosis)	67
	Mysidopsis intii	Neonate	28 d	NOEC (survival)	10
	Mysidopsis bahia	Larva	36 d	NOEC (reproduction)	61
	Tigriopus japonicus	Nauplius (24 h old)	20–30 d	EC10 (intrinsic rate of increase)	29.1
Echinoderm	*Dendraster excentricus*	Embryo	48 h	EC10 (abnormalities)	191
	Diadema antillarum	Larva	40 h	EC10 (abnormalities)	2.9
	Diadema savignyi	Gamete	48 h	NOEC (fertilisation and development)	23 [b]
	Paracentrotus lividus	Embryo	72 h	NOEC (larval survival)	50
	Strongylocentrotus purpuratus	Embryo	48 h	EC10 (abnormalities)	335
Gastropod	*Haliotis rufescens*	Embryo	14 d	NOEC (shell growth)	21.5
	Monodonta labio	Juvenile	30 d	EC10 (growth rate)	33.6
	Nassarius dorsatus	Larva	96 h	EC10 (growth rate)	64
Cnidarian (coral)	*Acropora digitifera*	Gamete	5 h	EC5 (fertilisation)	1 680
	Platygyra daedalea	Gamete	5 h	NOEC (fertilisation)	920
Cnidarian (sea anemone)	*Exaiptasia pulchella*	Adult	28 d	EC10 (reproduction)	65
Bivalve	*Crassostrea gigas*	Embryo	96 h	EC10 (reproduction)	431
	Mytilus galloprovincialis	Embryo	48 h	EC10 (survival)	270 [b]
	Mytilus trossolis	Embryo	48 h	EC20 (survival)	88

Taxonomic group	Species	Life stage	Duration	Toxicity measure [a] (endpoint)	Final toxicity value (µg/L)
Annelid	*Neanthes arenaceodentata*	Adult	90 d	EC10 (reproduction)	22.5
Fish	*Atherinops affinis*	Embryo	40 d	EC10 (larval survival)	3 600 [c]
	Cyprinidon variegatus	Juvenile	28 d	EC10 (growth)	20 300
	Oryzias melastigma	Juvenile	21 d	LC10 (mortality)	1 660

[a] The measure of toxicity being estimated/determined. EC5: 5% effect concentration; EC/LC10: 10% effect/lethal concentration; EC20: 20% effect concentration; NOEC: no-observed-effect concentration.
[b] Value is a geometric mean. See Appendix A for details.
[c] EC10 from DeForest and Schlekat (2013) using data supplied by authors.

4.2 Species sensitivity distribution

The cumulative frequency (species sensitivity) distribution (SSD) of the chronic marine toxicity data for nickel reported in Table 1 is shown in Figure 1. The SSD was plotted using the (shiny)ssdtools (V 0.4.0) software as described by Warne et al. 2025). The model provided a good fit to the data (Figure 1). Further details on the SSD fitting process are provided in Appendix D.

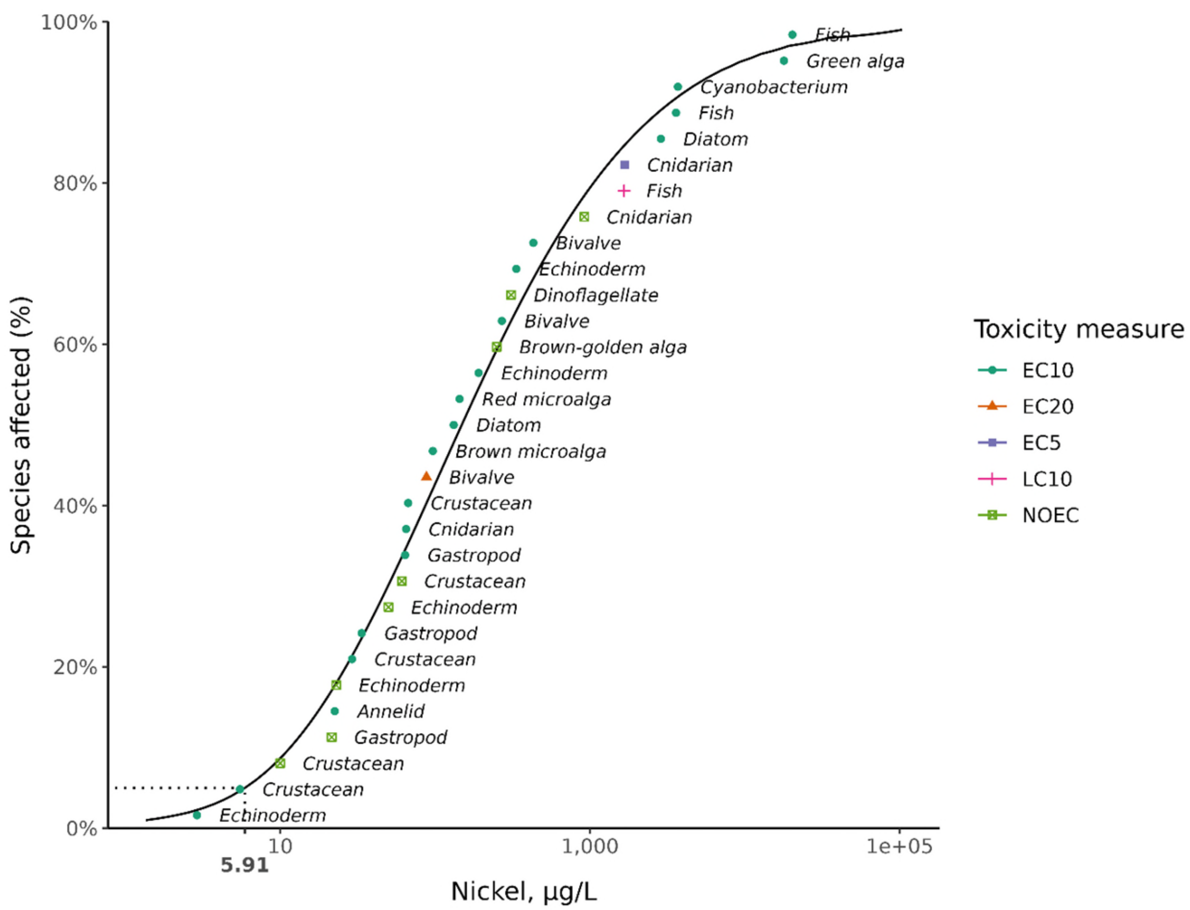


Figure 1 Species sensitivity distribution, dissolved nickel in marine water

Note: dotted line represents the interpolation of the 95% species protection value for nickel: i.e. 5.9 µg/L.

4.3 Default guideline values

It is important that the DGVs (Table 2) and associated information in this technical brief are used in accordance with the detailed guidance provided in the Australian and New Zealand Guidelines for Fresh and Marine Water Quality website (ANZG 2018).

The DGVs for nickel in marine water for 99%, 95%, 90% and 80% species protection are provided in Table 2. These supersede the ANZECC/ARMCANZ (2000a) DGVs for nickel in marine water. The DGVs relate to dissolved nickel, operationally defined as the < 0.45-µm filtered measurement. The 95% species protection DGV is protective of many species in the dataset; however, one echinoderm was not protected—even though other species in this taxonomic group was protected. The proportion of species not protected by the 95% species protection DGV is expected for the size of the dataset. The 95% species protection DGV of 5.9 µg/L is recommended for application to slightly-to-moderately disturbed ecosystems.

Table 2 Default guideline values, dissolved nickel in marine water, very high reliability

Level of species protection (% species)	DGV for dissolved nickel in marine water (µg/L) [a]
99	1.4
95	5.9
90	12
80	27

[a] The DGVs were derived using the (shiny)ssdtools (V 0.4.0) software. They have been rounded to two significant figures.

During the public consultation process, concerns were raised regarding the difficulty in the analytical detection of dissolved nickel concentrations below 5 µg/L in marine water samples. More importantly, erroneous results can be obtained if specialised trace metal sampling techniques are not followed. Sampling details are provided in the Australian guidelines for water quality monitoring and reporting (ANZECC/ARMCANZ 2000b). Analytical methods for trace nickel should be well within the scope of qualified analytical laboratories, and certified reference marine water samples are available for use in quality assurance. Where nickel speciation is a concern, available techniques include diffusive gradients in thin films (DGT) samplers (Zhang and Davison 1995) or the use of Chelex columns (Bowles et al 2006).

4.4 Reliability classification

The nickel in marine water DGVs have a very high reliability classification (Warne et al. 2018) based on the outcomes for the following three criteria:

- sample size—31 (preferred)
- type of toxicity data—chronic
- SSD model fit—good.

Glossary

Term	Definition
acute toxicity	A lethal or adverse sublethal effect that occurs as the result of a short exposure period to a chemical relative to the organism's life span.
benthic	Refers to organisms living in or on the sediments of aquatic habitats (e.g. lakes, rivers, ponds).
chronic toxicity	A lethal or sublethal adverse effect that occurs after exposure to a chemical for a period of time that is a substantial portion of the organism's life span or an adverse effect on a sensitive early life stage.
default guideline value (DGV)	A guideline value recommended for generic application in the absence of a more specific guideline value (e.g. site-specific guideline value), in the Australian and New Zealand Guidelines for Fresh and Marine Water Quality.
DOC	Dissolved organic carbon.
EC50 (median effective concentration)	The concentration of a substance in water or sediment that is estimated to produce a 50% change in the response being measured or a certain effect in 50% of the test organisms relative to the control response, under specified conditions.
ECx	The concentration of a substance in water or sediment that is estimated to produce an x% change in the response being measured or a certain effect in x% of the test organisms, under specified conditions.
endpoint	The specific response of an organism that is measured in a toxicity test (e.g. mortality, growth, a particular biomarker).
guideline value	A measurable quantity (e.g. concentration) or condition of an indicator for a specific community value below which (or above which, in the case of stressors such as pH, dissolved oxygen and many biodiversity responses) there is considered to be a low risk of unacceptable effects occurring to that community value. Guideline values for more than one indicator should be used simultaneously in a multiple lines of evidence approach.
humic substances	Organic substances only partially broken down that occur in water mainly in a colloidal state. Humic acids are large-molecule organic acids that dissolve in water.
LC50 (median lethal concentration)	The concentration of a substance in water or sediment that is estimated to be lethal to 50% of a group of test organisms, relative to the control response, under specified conditions.
LOEC (lowest observed effect concentration)	The lowest concentration of a material used in a toxicity test that has a statistically significant adverse effect on the exposed population of test organisms as compared with the controls.
MATC (maximum acceptable toxicant concentration)	The average (mean) of the NOEC and LOEC.
NOEC (no observed effect concentration)	The highest concentration of a material used in a toxicity test that has no statistically significant adverse effect on the exposed population of test organisms as compared with the controls.
species (biological)	A group of organisms that resemble each other to a greater degree than members of other groups and that form a reproductively isolated group that will not produce viable offspring if bred with members of another group.
species (chemical)	Most commonly used for metals, chemical species are different forms of a particular chemical that may include different oxidation states, isotopes, complexes with organic ligands (in the case of metals) or with particulate matter.
SSD (species sensitivity distribution)	A method that plots the cumulative frequency of species' sensitivities to a toxicant and fits a statistical distribution to the data. From the distribution, the concentration that should theoretically protect a selected percentage of species can be determined.

Term	Definition
toxicity	The inherent potential or capacity of a material to cause adverse effects in a living organism.
toxicity test	The means by which the toxicity of a chemical or other test material is determined. A toxicity test is used to measure the degree of response produced by exposure to a specific level of stimulus (or concentration of chemical) for a specified test period.

Appendix A: Toxicity data that passed the screening and quality assessment and were used to derive the default guideline values

Table A1 Summary, chronic toxicity data that passed the screening and quality assurance processes, nickel in marine water – temperate species

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentration (µg/L)	Reference
Diatom	*Skeletonema costatum*	–	96 h	EC10 (growth)	Seawater	20	28.5	8.4	142	Parametrix (2007g)
		–	96 h	EC10 (growth)	Seawater	20	30.2	8.3	89	Parametrix (2007g)
		–	96 h	EC10 (growth)	Seawater	20	29.4	8.2	383	Parametrix (2007g)
		–	96 h	EC10 (growth)	Seawater	20	29.2	8.3	190	Parametrix (2007g)
		–	96 h	EC10 (growth)	Seawater	20	29.4	8.3	43.5	Parametrix (2007g)
	–								**132**	**Value used in SSD** (geometric mean)
Green alga	*Dunaliella tertiolecta*	–	96 h	EC10 (growth)	Seawater	20	29.4	7.8	17 900	Parametrix (2007d)
	–								**17 900**	**Value used in SSD**
Red macroalga	*Champia parvula*	Adult	10 d	EC10 (reproduction)	Seawater	23	30	8	144	Parametrix (2007a)
	–								**144**	**Value used in SSD**
Brown macroalga	*Macrocystis pyrifera*	Zoospore	10 d	EC10 (germination)	Seawater	15	34	8	494	Golder (2007)
		Zoospore	10 d	EC10 (reproduction)	Seawater	15	34	8	96.7	Golder (2007)
	–								**96.7**	**Value used in SSD**
Crustacean	*Mysidopsis intii*	Neonate	28 d	NOEC (survival)	Seawater	20	34	–	10	Hunt et al. (2002)
		Neonate	28 d	EC10 (growth)	Seawater	20	34	–	45.2 [a]	Hunt et al. (2002)

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentration (µg/L)	Reference
	–								**10**	**Value used in SSD**
	Mysidopsis bahia	Larva	36 d	NOEC (reproduction)	Seawater	23	30	–	61	Gentile et al. (1982)
	–								**61**	**Value used in SSD**
Echinoderm	*Diadema antillarum*	Larva	40 h	EC10 (abnormalities)	Seawater	20	33	–	2.9 [a]	Bielmyer et al. (2005)
	–								**2.9**	**Value used in SSD**
	Paracentrotus lividus	Embryo	72 h	NOEC (egg production)	Seawater	18	35	–	500	Novelli et al. (2003)
		Embryo	72 h	NOEC (larval survival)	Seawater	18	35	–	50	Novelli et al. (2003)
	–								**50**	**Value used in SSD**
	Strongylocentrotus purpuratus	Embryo	48 h	EC10 (abnormalities)	Seawater	15.6	30	8.1	335	Parametrix (2007c)
	–								**335**	**Value used in SSD**
	Dendraster excentricus	Embryo	48 h	EC10 (abnormalities)	Seawater	15.4	30	8.1	191	Parametrix (2007c)
									191	**Value used in SSD**
Gastropod	*Haliotis rufescens*	Embryo	14 d	NOEC (shell growth)	Seawater	20	34	–	21.5	Hunt et al (2002)
		Embryo	14 d	EC10 (metamorphosis)	Seawater	20	34	–	36.4 [a]	Hunt et al (2002)
	–								**21.5**	**Value used in SSD**
Bivalve	*Crassostrea gigas*	Embryo	96 h	EC10 (reproduction)	Seawater	20.7	30	7.4	431	Parametrix (2007b)
	–								**431**	**Value used in SSD**
	Mytilus trossolis	Embryo	48 h	EC20 (survival)	Seawater	22–25	34	8	88	Nadella et al. (2009)
	–								**88**	**Value used in SSD**
	Mytilus galloprovincialis	Embryo	48 h	EC10 (survival)	Seawater	15.8	30	8.1	259	Parametrix (2007e)
		Embryo	48 h	EC10 (survival)	Seawater	16.1	30	7.9	228	Parametrix (2007e)
		Embryo	48 h	EC10 (survival)	Seawater	16	30	8.1	256	Parametrix (2007e)
		Embryo	48 h	EC10 (survival)	Seawater	16.1	30	8.1	350	Parametrix (2007e)

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentration (µg/L)	Reference
	–								**270**	**Value used in SSD** (geometric mean)
Annelid	*Neanthes arenaceodentata*	Adult	90 d	EC10 (reproduction)	Seawater	20	29.5	7.9	22.5	Parametrix (2007f)
	–								**22.5**	**Value used in SSD**
Fish	*Atherinops affinis*	Embryo	40 d	NOEC (larval survival)	Seawater	20	34	–	3 240	Hunt et al. (2002)
		Embryo	40 d	EC10 (larval survival)	Seawater	20	34	–	3 600 [a]	Hunt et al. (2002)
	–								**3 600**	**Value used in SSD**
	Cyprinidon variegatus	Juvenile	28 d	EC10 (growth)	Seawater	25	28–30	8.1	20 300	Golder (2007)
	–								**20 300**	**Value used in SSD**

[a] EC10 derived by from DeForest and Schlekat (2013).

Table A2 Summary, chronic toxicity data that passed the screening and quality assurance processes, nickel in marine water – tropical species

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentration (µg/L)	Reference
Cyanobacteria	*Cyanobium* sp.	6 x10^3 cells/mL	72 h	EC10 (growth rate)	Seawater with media	25	33	8	3 700	Alquezar and Anastasi (2013)
	–								**3 700**	**Value used in SSD**
Diatom	*Ceratoneis closterium* (G2 medium)	5–6 d old, 1–3 x 10^3 cells/mL	72 h	EC10 (growth rate)	Seawater	27	35	8.1	3 250	Gissi (2018)
	Ceratoneis closterium (F2 medium)	5–6 d old, 1–3 x 10^3 cells/mL	72 h	NOEC (growth rate)	Seawater	27	35	8.1	1 610	Gissi (2018)
		5–6 d old, 1–3 x 10^3 cells/mL	72 h	EC10 (growth rate)	Seawater	27	35	8.1	2 540	Gissi (2018)

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentra-tion (µg/L)	Reference
	–								**2 870**	**Value used in SSD** (geometric mean of EC10s)
Brown-golden alga	*Tisochrysis lutea*	5–6 d old, 1–3 x 10^3 cells/mL	72 h	NOEC (growth rate)	Seawater	27	35	8.1	250	Gissi (2018)
		5–6 d old, 1–3 x 10^3 cells/mL	72 h	EC10 (growth rate)	Seawater	27	35	8.1	330	Gissi (2018)
	–								**250**	**Value used in SSD**
Dinoflagellate	*Symbiodinium* sp. Freud. Clade C.	6–7 d old, 1–3 x 10^3 cells/mL	72 h	NOEC (growth rate)	Seawater	27	35	8.1	310	Gissi (2018)
	–								**310**	**Value used in SSD**
Crustacean	*Amphibalanus amphitrite*	Nauplius (<2 h old)	96 h	EC10 (metamorphosis)	Seawater	29	35	8.3	67	Gissi et al. (2018)
	–								**67**	**Value used in SSD**
	Acartia sinjiensis	Egg	80 h	EC10 (development)	Seawater	30	35	8.1	5.5	Gissi et al. (2018)
	–								**5.5**	**Value used in SSD**
	Tigriopus japonicus	Nauplius (<24 h old)	20–30 d	LC10 (mortality)	Artificial seawater	27	33	8.2	484	Wang et al. (2019)
		Nauplius (<24 h old)	20–30 d	NOEC (mortality)	Artificial seawater	27	33	8.2	99.8	Wang et al. (2019)
		Maturation stage	20–30 d	NOEC (mortality)	Artificial seawater	27	33	8.2	50.3	Wang et al. (2019)
		Maturation stage	20–30 d	LC10 (mortality)	Artificial seawater	27	33	8.2	43.9	Wang et al. (2019)
		Nauplius (<24 h old)	20–30 d	EC10 (intrinsic rate of increase [a])	Artificial seawater	27	33	8.2	29.1	Wang et al. (2019)
		Nauplius (<24 h old)	20–30 d	NOEC (intrinsic rate of increase)	Artificial seawater	27	33	8.2	50.3	Wang et al. (2019)
	–								**29.1**	**Value used in SSD**

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentration (µg/L)	Reference
Gastropod	*Nassarius dorsatus*	Larva (2 d old)	96 h	EC10 (growth rate)	Seawater	28	35	8.2	64	Gissi et al. (2018)
	–								**64**	**Value used in SSD**
	Monodonta labio	Juvenile (<10 d old)	30 d	LC10 (mortality)	Artificial seawater	27	33	8.2	57	Wang et al. (2019)
		Juvenile (<10 d old)	30 d	EC10 (growth rate)	Artificial seawater	27	33	8.2	33.6	Wang et al. (2019)
		Juvenile (<10 d old)	30 d	NOEC (growth rate)	Artificial seawater	27	33	8.2	21.7	Wang et al. (2019)
		Juvenile (<10 d old)	30 d	EC10 (shell length increment)	Artificial seawater	27	33	8.2	93.5	Wang et al. (2019)
		Juvenile (<10 d old)	30 d	NOEC (shell length increment)	Artificial seawater	27	33	8.2	53.9	Wang et al. (2019)
	–								**33.6**	**Value used in SSD**
Cnidarian (coral)	*Acropora digitifera*	Gamete	5 h	NOEC (fertilisation)	Seawater	25	34	8.1	940	Gissi et al. (2017)
		Gamete	5 h	EC10 (fertilisation)	Seawater	25	34	8.1	2 000	Gissi et al. (2017)
		Gamete	5 h	EC5 (fertilisation)	Seawater	25	34	8.1	1 680	Gissi et al. (2017)
	–								**1 680**	**Value used in SSD**
	Platygra daedalea	Gamete	5 h	NOEC (fertilisation)	Seawater	25	34	8.1	920	Gissi et al. (2017)
	–								**920**	**Value used in SSD**
Cnidarian (sea anemone)	*Exaiptasia pulchella*	Lacerate tentacle	14 d	EC10 (development)	Seawater	25	34	8.2	260	Howe et al. (2014)
		Adult	28 d	EC10 (reproduction—total number of offspring)	Seawater	25	–	8.2	260	Howe et al. (2014)
		Adult	28 d	EC10 (reproduction—total number of juveniles)	Seawater	25	–	8.2	65	Howe et al. (2014)
	–								**65**	**Value used in SSD**

Taxonomic group	Species	Life stage	Duration	Toxicity measure (test endpoint)	Test medium	Temp. (°C)	Salinity (‰)	pH	Concentra-tion (µg/L)	Reference
Echinoderm	*Diadema savignyi*	Gamete	48 h	NOEC (fertilisation and development)	Seawater	25	34	8.1	23.5	Rosen et al. (2015)
		Gamete	48 h	NOEC (fertilisation and development)	Seawater	25	34	8.1	22.5	Rosen et al. (2015)
	–								**23**	**Value used in SSD** (geometric mean)
Fish	*Oryzias melastigma*	Juvenile (1 month post hatching)	21 d	LC10 (mortality)	Artificial seawater	27	30	8.94–8.98	1 660	Wang et al. (2019)
	–								**1 660**	**Value used in SSD**

[a] Intrinsic rate of increase: population growth = number of births – number of deaths.

Appendix B: Assessment of specific data selection decisions

Converted chronic values

The acceptable quality dataset included nine chronic toxicity values that needed to be converted to negligible effect equivalents if they were going to be used for the derivation, as follows:

- 42-d MATC for the crustacean *Portunus pelagicus* (Mortimer and Miller 1994) – 32 µg/L
- 10-d LOEC for the crustacean *Acartia pacifica* (Mohammed et al. 2010) – 50 µg/L
- 48-h EC50 for the crustacean *Artemia salina* (Kissa et al. 1984) – 4660 µg/L
- 30-d EC50 value for the crustacean *Litopenaeus vannamei* (Leonard et al. 2011) – 445 µg/L
- 30-d EC50 value for the crustacean *Excirolana armata* (Leonard et al. 2011) – 1350 µg/L
- 96-h EC50 for the echinoderm *Evichinus chloroticus* (Blewett et al., 2016) – 14 µg/L
- 96-h EC50 for the bivalve *Mytilus edulis* (Martin et al., 1981) – 890 µg/L
- 20-d EC50 for the annelid *Hydroides elegans* (Gopalakrishnan et al. 2008) – 160 µg/L
- 64-h EC50 for the echinoderm *Hemicentrotus pulcherrimus* (Hwang et al. 2012) – 34 µg/L

An SSD with these values included in the dataset following conversion to negligible effect equivalents using the default factors in Warne et al. (2018) is shown in Figure B1, with the resulting PC values shown in Table B1 (dataset A). The PC values were slightly lower than the DGVs (dataset D in Table B1); however, the greater uncertainty associated with the need to convert the above values to negligible effect equivalents using default factors drove the decision to exclude these values.

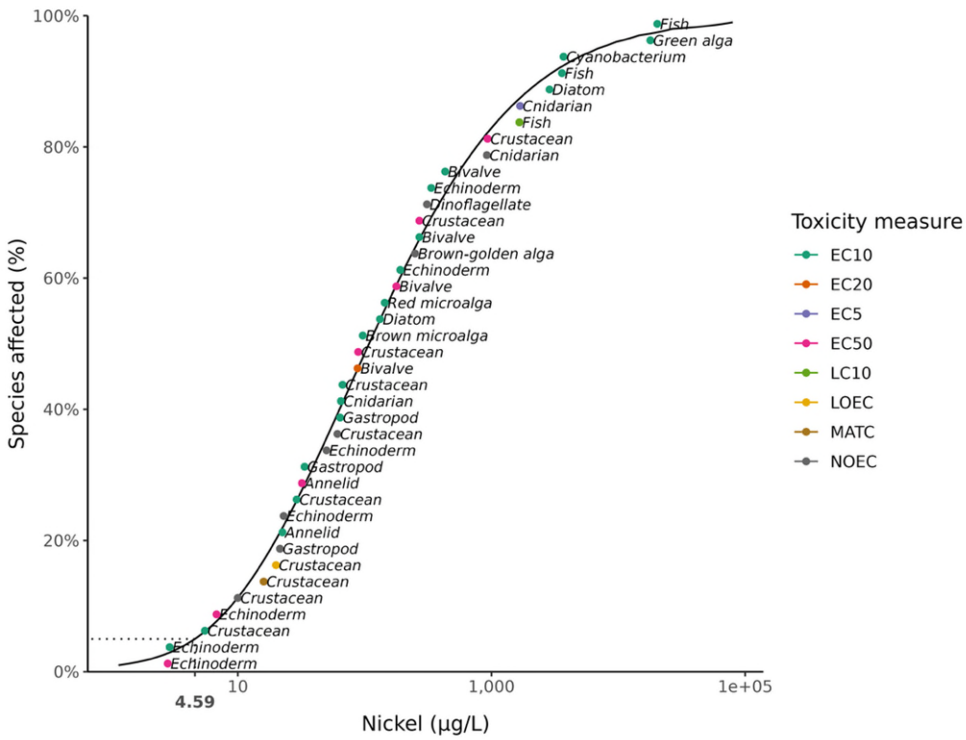


Figure B1 Species sensitivity distribution for nickel in marine water using all acceptable data (excluding two "<" values) (n = 40)

Table B1 Protective concentrations for various nickel marine toxicity datasets

Level of species protection (% species)	Protective concentration (µg/L) [a]			
	A. Acceptable dataset excluding two "<" values (n = 40)	B. Acceptable dataset excluding MATC, LOEC and EC50 values (n = 33)	C. Acceptable dataset excluding NOEC and EC20 values (n = 22)	D. Final dataset [b] (n = 31)
99	1.2	1.4	0.65	1.4
95	4.6	5.7	4.9	5.9
90	8.8	11	12	12
80	20	25	32	27

[a] Protective concentration values were derived using the (shiny)ssdtools (V 0.4.0) software and were rounded to two significant figures.
[b] Dataset used to derive the DGVs.

NOECs reported as "less than" values

Warne et al. (2018) recommends that "<" (or "≤") values should be excluded unless: (i) there are no other data for the species, (ii) the data point sits at the lower end of the distribution of species sensitivities, or (iii) the exclusion of the data would result in a less conservative GV. The two < values of <280 µg/L for *A. aspera* (Gissi et al. 2017) and <10 µg/L for *H. pulcherrimus* (Hwang et al. 2012) were the only values for these species. The value of for *H. pulcherrimus* was within the lowest 10% of the full dataset, while the value for *A. aspera* was in the upper half of the dataset, although was the lowest value for a coral in the dataset. However, inclusion of the two values had very little effect on the estimated PC values, as shown in Table B1 (dataset B), relative to the final dataset (dataset D) (SSD not shown). The PC differed by between 0% and 8% from the DGVs based on the final dataset. Given the lack of influence of the values on the DGVs, and the higher uncertainty associated with "<" values, the two values for *A. aspera* and *H. pulcherrimus* were excluded from the final dataset.

NOECs and EC20

The acceptable quality dataset included eight NOECs and an EC20. Such values are considered as 'non-preferred' by Warne et al. (2018) and their inclusion for datasets with more than eight values needs to be justified. An SSD with these values excluded from the dataset is shown in Figure B2, with the resulting PC values shown in Table B1 (dataset C). The exclusion of the NOEC and EC20 values did not have a large effect on the DGVs relative to the dataset with the NOEC and EC20 values included (Table B1, dataset D), although it did reduce the PC99 and PC95 values by approximately 50% and 20%, respectively. However, five of the NOEC values are within the 10 lowest values in the final dataset (Figure 1), and thus, help to populate the lower end of the SSD, including filling in a gap around the 10th percentile, between the crustacean and the annelid (Figure B2). Moreover, the inclusion of the NOEC and EC20 values adds nine additional species and two additional taxonomic groups (golden alga, dinoflagellate) to the dataset, thus, markedly increasing the taxonomic representation. Overall, the reliability of the PC values is not affected by the inclusion or exclusion of the NOEC and EC20 values, being assigned as very high reliability wither way. Finally, and as described in Section 4.1, it is also important to note that there were several instances where the application of professional judgement led to a decision that an available NOEC was a more appropriate estimate than an available EC10. Consequently, on balance, it was considered more appropriate to include the NOEC and EC20 values in the final dataset.

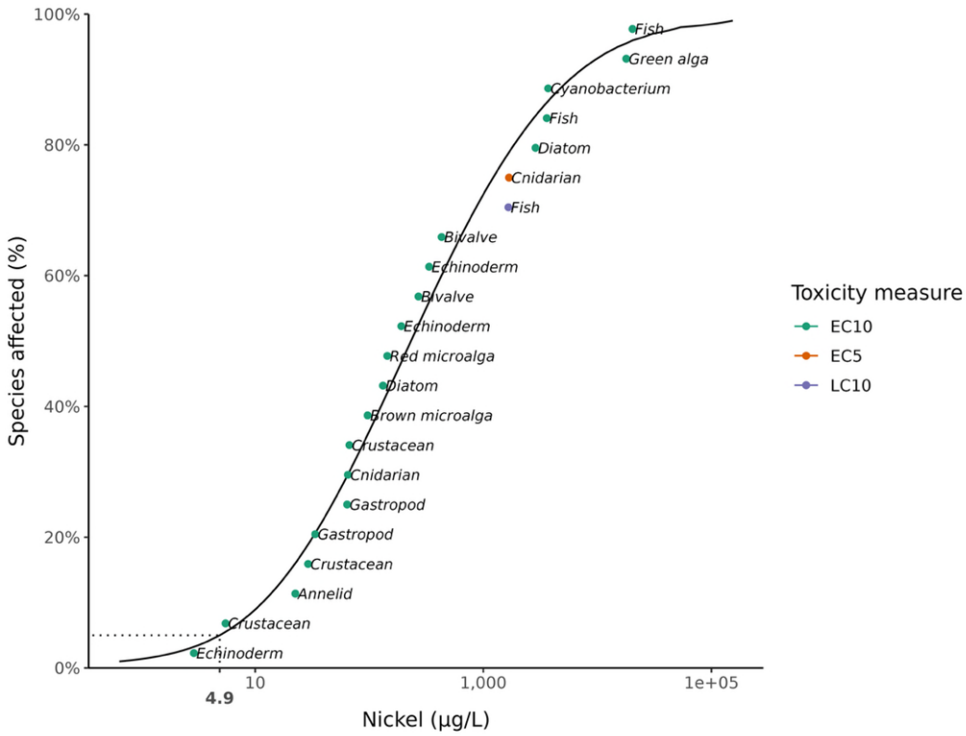


Figure B2 Species sensitivity distribution for nickel in marine water using only the EC5, EC10 and LC10 values (n = 22).

Appendix C: Modality assessment for nickel toxicity to marine species

A modality assessment was undertaken for the nickel in marine water toxicity dataset according to the four questions stipulated in Warne et al. (2018). These questions and their answers are as follows.

1. Is there a specific mode of action that could result in taxa-specific sensitivity?

There are limited studies on the mechanism of nickel toxicity to marine organisms. Nickel possibly disrupts ion regulatory balance in invertebrates and could be a respiratory toxicant to fish. Nickel can also be a micronutrient for plants. It is likely that there are taxa-specific modes of action for nickel toxicity, although evidence is limited.

2. Do the data suggest bimodality?

Visual representation of the data, calculation of the bimodality coefficient (BC), and consideration of the range in the effect concentrations are recommended lines of evidence for evaluating whether bimodality or multimodality of the dataset is apparent. For this assessment:

- the histogram of the log10-tranformed nickel marine toxicity data (Figure C 1) indicates that the data are normally distributed and unimodal
- data that span large ranges (>4 orders of magnitude) indicate potential for underlying bimodality or multimodality (Warne et al. 2018); the nickel data span less than 4 orders of magnitude
- when the BC is greater than 0.555, it indicates that the data do not follow a typical normal distribution and may be bimodal; the BC for the log transformed data is 0.354 and, therefore, is not indicative of bimodality.

Based on the lines of evidence described above, the nickel marine toxicity dataset does not appear to be bimodal.

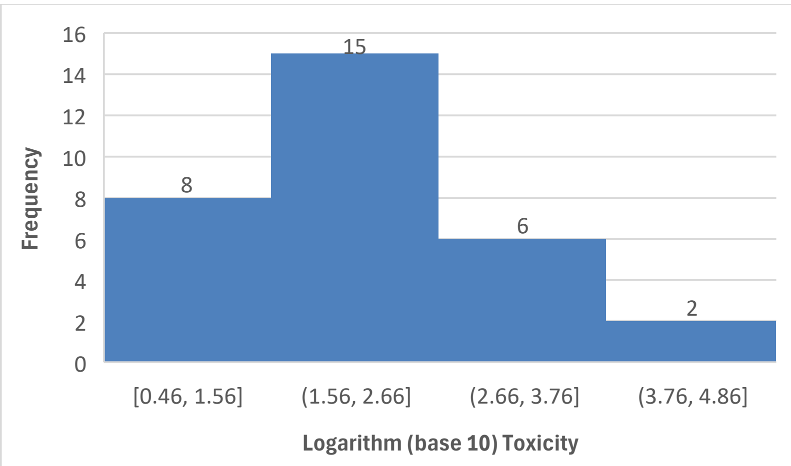
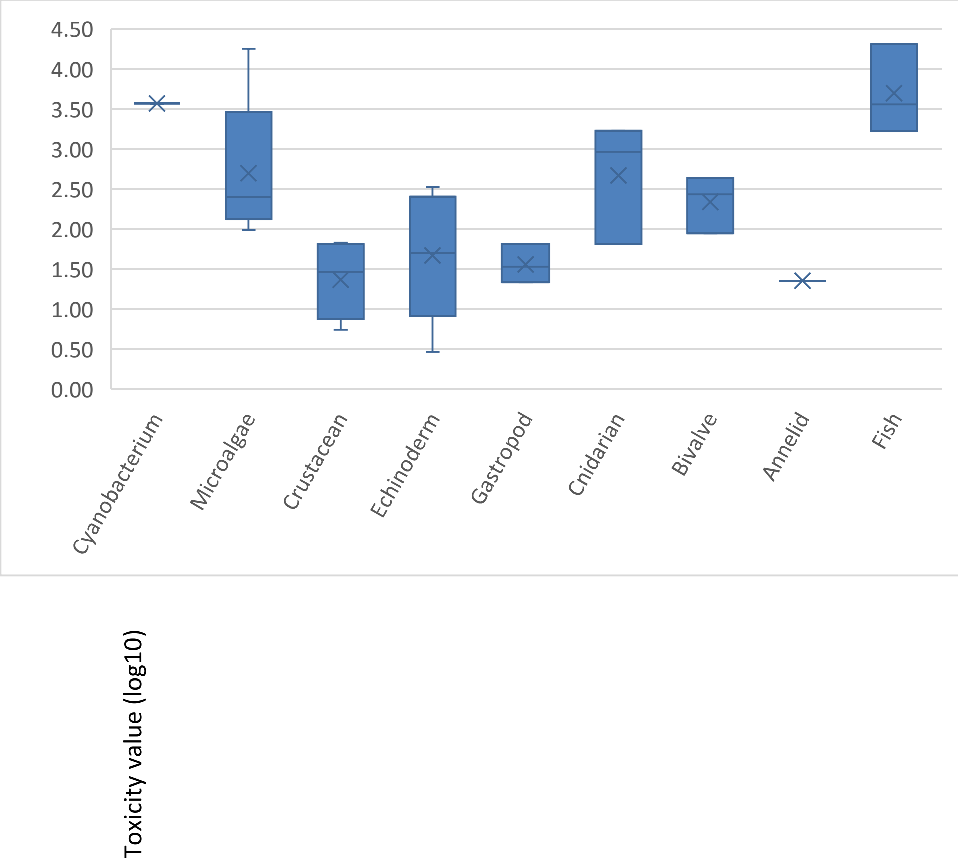


Figure C1 Histogram, log10-transformed nickel marine toxicity data



Nickel in marine water

lefinitive conclusions.
nd annelids (polychaetes)
ria (n = 3) and macroalgae
e (n = 7) being least sensitive
sensitivity, particularly the
itivity (Figure B2).

Figure C2 Box plots, log10-transformed nickel marine toxicity data for individual phyla

4. Is it likely that indications of bimodality or multimodality or distinct clustering of taxa groups are not due to artefacts of data selection, small sample size, test procedures, or other reasons unrelated to a specific mode of action?

It is not possible to determine if indications of taxa-specific sensitivity are real or due to artefacts. Regardless, the dataset displays no indications of bimodality or multimodality and, therefore, the full dataset was used for the derivation of the DGVs.

Appendix D: Details of the shinyssdtools SSD fitting process for the nickel marine toxicity dataset

The SSD software package shinyssdtools (V 0.4.0) uses a model averaging approach to generating a SSD (as described by Fox et al. 2021). Figure D 1 shows the full set of default distributions fitted to the nickel marine toxicity dataset, while Table D 1 provides the relevant goodness of fit and weighting estimates for each of the distributions. The final, model-averaged SSD is shown in Figure 1 of the main report.

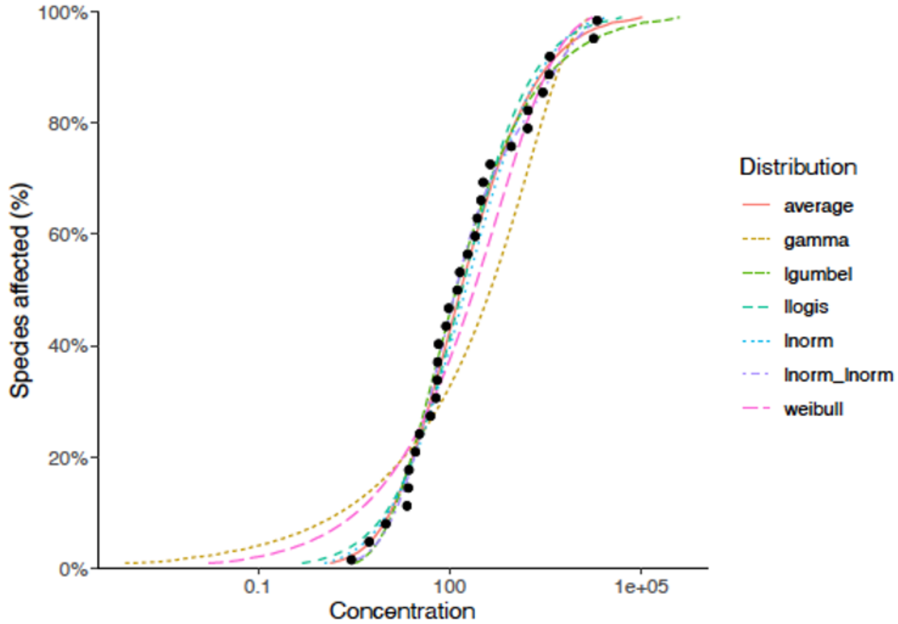


Figure D 1 SSDs for set of six default distributions used to model the nickel marine toxicity dataset

Table D 1 Goodness of fit and weighting estimates for six default distributions used in model averaging for the nickel marine toxicity dataset

Distribution	Akaike's Information Criterion (AIC)	Corrected Akaike's Information Criterion (AICc)	Delta	Weight
Log gumbel	461	461	0	0.407
Log normal	461	462	0.353	0.342
Log logistic	462	463	1.300	0.212
Log normal - Log normal	464	467	5.260	0.029
Weibull	468	469	7.600	0.009
Gamma	477	477	16.000	0.000

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